LECTURE NOTES - SCANNED COPY

Lecture 1: Introduction to Artificial Neural Networks

An artificial neural network (ANN) is a computational model inspired by biological neural networks.

Neurons receive input signals, apply weights, add a bias term, and pass through an activation function.

Lecture 2: Perceptron Model and Linear Separability

The perceptron is a binary classification algorithm invented by Frank Rosenblatt in 1957.

It computes a weighted sum of inputs and outputs 1 if the sum exceeds a threshold, otherwise 0.

Lecture 3: Multilayer Perceptrons and Activation Functions

Multilayer perceptrons (MLP) contain one or more hidden layers between the input and output layers.

Common activation functions include Sigmoid, Hyperbolic Tangent (Tanh), and Rectified Linear Unit (ReLU).

Lecture 4: Backpropagation and Gradient Descent

Backpropagation is an algorithm used to calculate the gradient of the loss function with respect to weights.

Gradient descent iteratively adjusts network weights in the direction of steepest descent to minimize loss.

Lecture 5: Loss Functions and Optimization

Mean Squared Error (MSE) is commonly used for regression tasks.

Cross-Entropy Loss is the standard objective function for multi-class classification problems.

Lecture 6: Overfitting and Regularization Techniques

Overfitting occurs when a neural network models training noise rather than the underlying distribution.

Regularization techniques include L1 (Lasso), L2 (Ridge / Weight Decay), and Dropout.

Lecture 7: Convolutional Neural Networks (CNN)

CNNs are specialized neural networks for processing grid-structured data such as images.

Core operations include convolution filters, pooling layers (max pooling), and fully connected layers.

Lecture 8: Recurrent Neural Networks (RNN)

RNNs process sequential data by maintaining an internal hidden state across time steps.

Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) mitigate the vanishing gradient problem.

Lecture 9: Attention Mechanisms and Transformers

The attention mechanism allows models to focus dynamically on different segments of input sequences.

The Transformer architecture relies entirely on self-attention mechanisms without recurrence.

Lecture 10: Deep Learning Optimization and Schedulers

Adaptive optimizers like Adam and RMSprop adjust learning rates per parameter.

Learning rate warmup and cosine annealing schedules improve model generalization.

Lecture 11: Autoencoders and Latent Representation

Autoencoders are unsupervised neural networks designed to learn efficient codings of input data.

They consist of an encoder mapping input to code, and a decoder reconstructing input from code.

Lecture 12: Generative Adversarial Networks (GAN)

GANs consist of two competing networks: a generator synthesizing data and a discriminator evaluating realism.

They are trained simultaneously in a minimax game framework.